

2. Assume that the weight of the beam and empty box is negligible compared to the other forces.

Balance forces in the x and y directions, (refer to free body diagram above)

$$R_x = T \cos 50^\circ = 2600 \cos 50^\circ = 1671 \text{ N}$$

$$R_y = W_1 + W_2 + T \sin 50^\circ = 250 + 610 \times 9.81 + T \sin 50^\circ = 8230 \text{ N}$$

$$R = (1671^2 + 8230^2)^{1/2} \text{ (use of pythagorus theorem)}$$

$$= 8400 \text{ N}$$

C1

C1

C1

A1

- 2 (a) Newton's second law of motion states that the rate of change of momentum of a body is directly proportional to the resultant force acting on it and the change takes place in the direction of the resultant force.

B1

- (b) (i) Since the momentum of a photon  $p = \frac{h}{\lambda}$  and energy  $E = \frac{hc}{\lambda}$ ,  $p = \frac{E}{c}$

M1

Taking the initial direction as positive,

$$\text{Change in momentum of photon} = p_f - p_i = -\frac{E}{c} - \frac{E}{c} = -\frac{2E}{c}$$

M1

By conservation of momentum, change in momentum of spacecraft =  
- change in momentum of photon =  $\frac{2E}{c}$ .

A1

Note: photons do not have mass, so do not use  $p = mc$  or  $E = \frac{1}{2}mc^2$

- (ii) Since  $P$  is the rate of energy received per unit time,  $P = NE / t$   
where  $N$  is the number of photons reflected.

M1

Average rate at which photons are reflected is given by  $\frac{N}{t} = \frac{P}{E}$

$$\begin{aligned}
 \text{Thrust force} &= \text{rate of change of momentum} \\
 &= \text{rate of photons incident on sail} \times \text{change in momentum} \\
 &= \frac{P}{E} \times \frac{2E}{c} = \frac{2P}{c} = \frac{2(1400)}{3.0 \times 10^8} \\
 &= 9.3 \times 10^{-6} \text{ N}
 \end{aligned}$$

M1

A1

(iii)  $F = ma$   
 $10^6 \times 9.3 \times 10^{-6} = 1000 \text{ a}$   
 $a = 9.3 \times 10^{-3} \text{ m s}^{-2}$

A1

- (iv) The change in momentum for each photon will be halved as the final momentum is zero.  
 Hence, the thrust and acceleration of the spacecraft are also halved.

B1

B1